

DSAA3073: Theories in Data Science

Week 9 Open Investigation

When Does Multiplicative Weights Learn?

A load-forecasting decision under altered information structures

Basic analysis. Combining electricity-demand forecasts

Use the full-information expert-advice protocol and linear MWU from the Week 9 Learning Sheet

A regional grid operator purchases balancing reserve every fifteen minutes. Four forecasting services supply demand predictions. Before actual demand is known, the operator forms a distribution over the services; after demand is observed, the normalized absolute error of every service is available. Thus each round reveals a full loss vector in $[0, 1]^4$.

At the beginning of one representative round the weights are $w_t = (4, 3, 2, 1)$ and the revealed loss vector is $\ell_t = (0.10, 0.40, 0.80, 0.30)$. The operational implementation charges the distribution-weighted loss rather than sampling a single service. For the one-round audit use $\eta = 0.20$.

Basic Task 1. Execute and audit one linear MWU round

Compute p_t , the operator's mixture loss, every updated weight, and the new total potential. Verify the exact potential identity. Explain which pieces of the calculation use bounded losses, full feedback, and commitment before the current loss vector is revealed.

Basic analysis. Translate the theorem into an operational guarantee

Regret compares with one fixed service on the same realized sequence

The operator plans a horizon of $T = 400$ rounds and initializes all four experts equally. Assume the standard Week 9 protocol throughout the horizon.

Basic Task 2. Tune the rate and interpret the regret bound

State the linear MWU comparator theorem and its $[0, 1]$ external-regret corollary. Select the legal fixed-horizon rate, calculate the numerical bound, and translate it into average excess loss per round. State what this conclusion does and does not say about the best service at each individual time.

Thinking investigation. Formalize the information contract

The loss sequence may be adaptive without observing the current private action

Write the online problem as

$$(\mathcal{I}_t, p_t, \ell_t, \mathcal{C}, \mathcal{G}),$$

where $\mathcal{I}_t$ is the information available when p_t is chosen, ℓ_t is the current loss vector, $\mathcal{C}$ is the fixed-expert comparator, and $\mathcal{G}$ is the regret guarantee. The Week 9 theorem permits dependence on the past, but the current loss must not be chosen after observing the learner's current private draw.

Investigation 1. Let the adversary observe the sampled service

Suppose the operator samples I_t from p_t , and a strategic market actor observes I_t before assigning losses. On every round it sets $\ell_t(I_t) = 1$ and $\ell_t(i) = 0$ for $i \neq I_t$. Does the expected sampled-loss interpretation of the Week 9 theorem remain valid? Derive a pathwise lower bound on external regret after T rounds for n services.

Thinking investigation. Delay the full-information feedback

A block protocol yields a new analytic rate

Now restore the non-anticipating environment, but suppose complete loss vectors arrive only at the end of blocks of B rounds. During a block the operator must keep one distribution fixed. Assume B divides T and define the block-average loss

$$\bar{\ell}_s = \frac{1}{B} \sum_{t=(s-1)B+1}^{sB} \ell_t \in [0, 1]^n, \quad s = 1, \dots, \frac{T}{B}.$$

Investigation 2. Recover a valid regret guarantee under batching

Explain why the original round-by-round implementation is unavailable. Apply linear MWU to the block-average vectors and derive an upper bound on the original, unaveraged T -round regret. Optimize the learning rate in the uncapped regime and identify the exact price of delayed updates.

Synthesis. State the assumption-guarantee map

Compare immediate full feedback with mixture loss, current-action-revealing feedback with sampled loss, and non-anticipating block feedback. For each regime, state the valid comparator and the strongest analytic conclusion.